

MINIMAL and ORANI-G Model

A Prelude to TERM Model

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CES Production Function Review

General CES production function: $Q = \left\{ \sum_i \delta_i X_i^{-\rho} \right\}^{-\frac{1}{\rho}}$

or $Q = \left\{ \sum_i \delta_i X_i^{\frac{\sigma-1}{\sigma}} \right\}^{\frac{\sigma}{\sigma-1}}$ where $\sigma = \frac{1}{1+\rho}$ is elasticity of substitution and $0 < \sigma < \infty$, For convenience, let's simplify this to $Q(\mathbf{X}_i; \sigma_q)$. Where $\mathbf{X}_i = X_1, X_2, \dots, X_i, \dots, X_n$

So, for example: $X = \left(\delta_K \cdot K^{\frac{\sigma_X-1}{\sigma_X}} + \delta_L \cdot L^{\frac{\sigma_X-1}{\sigma_X}} \right)^{\frac{\sigma_X}{\sigma_X-1}}$

can be written as $X(K, L; \sigma_X)$.

Remember also that:

- $X(K, L; 0)$ is Leontief production function, and
- $X(K, L; 1)$ is Cobb-Douglas production function.

CES Production Function with Technological Change

CES production fn. with technological change: $Q = \frac{1}{A_Q} \left\{ \sum_i \delta_i \left(\frac{X_i}{A_i} \right)^{-\rho} \right\}^{-\frac{1}{\rho}}$

or $Q = \frac{1}{A_Q} \left\{ \sum_i \delta_i \left(\frac{X_i}{A_i} \right)^{\frac{\sigma-1}{\sigma}} \right\}^{\frac{\sigma}{\sigma-1}}$ where $0 < \sigma = \frac{1}{1+\rho} < \infty$.

We can simplify this to $Q \left(\frac{\mathbf{X}_i}{\mathbf{A}_i}; \sigma_Q, A_Q \right)$.

So, for example: $X = \frac{1}{A_X} \left(\delta_K \cdot \frac{K}{A_K}^{\frac{\sigma_X-1}{\sigma_X}} + \delta_L \cdot \frac{L}{A_L}^{\frac{\sigma_X-1}{\sigma_X}} \right)^{\frac{\sigma_X}{\sigma_X-1}}$

can be written as $X \left(\frac{K}{A_K}, \frac{L}{A_L}; \sigma_X, A_X \right)$.

Then:

- $X \left(\frac{K}{A_K}, \frac{L}{A_L}; 0, A_X \right)$ is Leontief production function, and
- $X \left(\frac{K}{A_K}, \frac{L}{A_L}; 1, A_X \right)$ is Cobb-Douglas production function.

Database structure of 2x2 Model

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- I-O Table of the island's economy

	Industry		Consumption	Total
	1-corn	2-meat		
1-corn			100	100
2-meat			200	200
Salary/labor cost	40	140		180
Surplus/land rental	60	60		120
Total	100	200	300	300

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Database structure of MINIMAL Model

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		Absorption Matrix					
		1	2	3	4	5	
		Producers	Investors	Household	Export	Government	Total Sales
	Size	← I →	← 1 →	← 1 →	← 1 →	← 1 →	
Domestic Flows	↑ C ↓	USE(commodity,"dom",user)					
Imported Flows	↑ C ↓	USE(commodity,"imp",user)					
Labour	↑ 1 ↓	FACTOR (labour)	C = Number of Commodities = 7 I = Number of Industries = 7				
Capital	↑ 1 ↓	FACTOR (capital)					
Output tax	↑ 1 ↓	V1PTX					

		Tax on Imports
Size	← 1 →	
↑ C ↓		V0MTX

Database structure of ORANI-G Model

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		Absorption Matrix					
		1	2	3	4	5	6
		Producers	Investors	Household	Export	Government	Change in Inventories
	Size	← I →	← I →	← 1 →	← 1 →	← 1 →	← 1 →
Basic Flows	↑ C×S ↓	V1BAS	V2BAS	V3BAS	V4BAS	V5BAS	V6BAS
Margins	↑ C×S×M ↓	V1MAR	V2MAR	V3MAR	V4MAR	V5MAR	n/a
Taxes	↑ C×S ↓	V1TAX	V2TAX	V3TAX	V4TAX	V5TAX	n/a
Labour	↑ O ↓	V1LAB	C = Number of Commodities I = Number of Industries S = 2: Domestic, Imported, O = Number of Occupation Types M = Number of Commodities used as Margins				
Capital	↑ 1 ↓	V1CAP					
Land	↑ 1 ↓	V1LND					
Production Tax	↑ 1 ↓	V1PTX					
Other Costs	↑ 1 ↓	V1OCT					

Joint Production Matrix	
Size	← I →
↑ C ↓	MAKE

Import Duty	
Size	← 1 →
↑ C ↓	V0TAR

2x2 vs MINIMAL: Production

2x2 model production function:

$$X_i(K_i, L_i; \sigma_X)$$

With optimization and linearization:

- $k_i = x_i - \sigma_X (r - c_i)$
- $l_i = x_i - \sigma_X (w - c_i)$
- $c_i = S_i^L w + S_i^K r$

Now, define Z_{ci} intermediate input commodity c used in production of X_i :

MINIMAL model's CES *separable* production function:

$$X_i(V_i(K_i, L_i; \sigma_V, A_V), \mathbf{Z}_{ci}; 0)$$

With optimization and linearization:

- $k_i = v_i - \sigma_V (r_i - c_i)$
- $l_i = v_i - \sigma_V (w_i - c_i)$
- $c_i = S_K \cdot r_i + S_L \cdot w$
- $v_i = a_v + x_i$
- $Z_{ci} = x_i$

ORANI-G vs MINIMAL (separable) Production Function

MINIMAL and ORANI-G Model

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MINIMAL production function:

$$X_i(V_i(K_i, L_i; \sigma_V, A_V), \mathbf{Z}_{ci}; 0)$$

ORANI-G production function:

$$X_i \left(V_i \left(\frac{K_i}{A_i^K}, L_i \left(\mathbf{L}_{io}^S; \sigma_L, A_i^L \right), \frac{N_i}{A_i^N}; \sigma_V, A_i^V \right), \frac{\mathbf{Z}_{ci}}{\mathbf{A}_{ci}^Z}; 0, A_i^X \right)$$

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ORANI-G: Production

ORANI-G model's CES *separable* production function:

$$X_i \left(V_i \left(\frac{K_i}{A_i^K}, L_i \left(\mathbf{L}_{io}^S; \sigma_L, A_i^L \right), \frac{N_i}{A_i^N}; \sigma_V, A_i^V \right), \frac{Z_{ci}}{\mathbf{A}_{ci}^Z}; 0, A_i^X \right)$$

With optimization and linearization:

- $k_i - a_i^K = v_i - \sigma_V (r_i + a_i^K - c_i)$
- $l_i - a_i^L = v_i - \sigma_V (w_i + a_i^L - c_i)$
- $n_i - a_i^N = v_i - \sigma_V (d_i + a_i^N - c_i)$
- $c_i = S_K \cdot r_i + S_L \cdot w + S_N \cdot d$
- $v_i - (a_i^V + a_i^X) = x_i$
- $z_{ci} - (a_{ci}^Z + a_i^X) = x_i$
- $l_{io}^S = l_i - \sigma_L (w_{io}^S - w_i)$
- $w_i = \sum_o S_{io}^{LS} w_{io}^S$

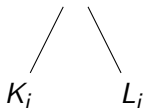
2x2 vs MINIMAL: Nested Production Function

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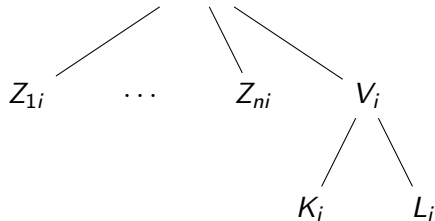
2x2 Model:

$$X_i(K_i, L_i; \sigma_X)$$



MINIMAL model:

$$X_i(V_i(K_i, L_i; \sigma_V, A_V), \mathbf{Z}_{ci}; 0)$$



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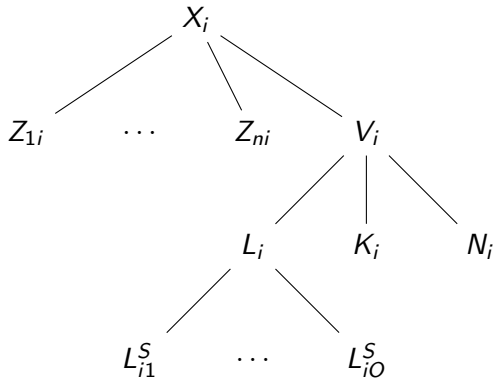
ORANI-G vs MINIMAL Model: Nested Production Function

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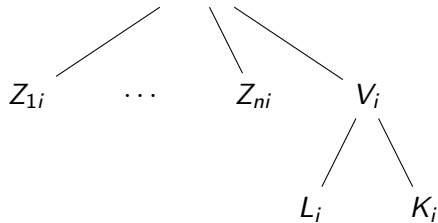
ORANI-G model:

$$X_i \left(V_i \left(\frac{K_i}{A_i^K}, L_i \left(\mathbf{L}_{i0}^S; \sigma_L, A_i^L \right), \frac{N_i}{A_i^N}; \sigma_V, A_i^V \right), \frac{Z_{ci}}{A_i^Z}; 0, A_i^X \right)$$



MINIMAL model:

$$X_i(V_i(K_i, L_i; \sigma_V, A_V), Z_{ci}; 0)$$



MINIMAL Production Function Tablo Equations

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Equation E_x1lab

$$(all,i,IND) \ x1lab(i) = x1prim(i) - SIGMA1PRIM(i)*[p1lab-p1prim(i)];$$

Equation E_x1cap

$$(all,i,IND) \ x1cap(i) = x1prim(i) - SIGMA1PRIM(i)*[p1cap(i)-p1prim(i)];$$

Equation E_p1prim

$$(all,i,IND) \ V1PRIM(i)*p1prim(i) \\ = FACTOR("Labour",i)*p1lab + FACTOR("Capital",i)*p1cap(i);$$

Equation E_x1 # demand for commodity composites #

$$(all,c,COM)(all,i,IND) \ x_s(c,i)= x1tot(i);$$

Equation E_x1prim # demand for primary-factor composites #

$$(all,i,IND) \ x1prim(i) = a1prim(i) + x1tot(i);$$

Production Function Tablo Equations (ORANI-G) - 1

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E_x1lab_o # Industry demands for effective labour #

```
(all,i,IND) x1lab_o(i) - a1lab_o(i) =  
x1prim(i) - SIGMA1PRIM(i)*[p1lab_o(i) + a1lab_o(i) - p1prim(i)];
```

E_p1cap # Industry demands for capital #

```
(all,i,IND) x1cap(i) - a1cap(i) =  
x1prim(i) - SIGMA1PRIM(i)*[p1cap(i) + a1cap(i) - p1prim(i)];
```

E_p1lnd # Industry demands for Land #

```
(all,i,IND) x1lnd(i) - a1lnd(i) =  
x1prim(i) - SIGMA1PRIM(i)*[p1lnd(i) + a1lnd(i) - p1prim(i)];
```

E_p1prim # Effective price term for factor demand equations #

```
(all,i,IND) V1PRIM(i)*p1prim(i) = V1LAB_O(i)*[p1lab_o(i) + a1lab_o(i)]  
+ V1CAP(i)*[p1cap(i) + a1cap(i)] + V1LND(i)*[p1lnd(i) + a1lnd(i)];
```

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```
E_x1_s # Demands for commodity composites #
(all,c,COM)(all,i,IND)  x1_s(c,i) - [a1_s(c,i) + a1tot(i)] = x1tot(i);

E_x1prim # Demands for primary factor composite #
(all,i,IND)  x1prim(i) - [a1prim(i) + a1tot(i)] = x1tot(i);

E_x1lab # Demand for labour by industry and skill group #
(all,i,IND)(all,o,OCC)
  x1lab(i,o) = x1lab_o(i) - SIGMA1LAB(i)*[p1lab(i,o) - p1lab_o(i)];

E_p1lab_o # Price to each industry of labour composite #
(all,i,IND) [TINY+V1LAB_O(i)]*p1lab_o(i)
              = sum{o,OCC, V1LAB(i,o)*p1lab(i,o)};
```

MINIMAL & ORANI-G Demand for Commodities

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MINIMAL distinguishes commodities by its origin (local Q_c^D or import Q_c^M) and demand come from more than one entity. Total demand for local commodities: Demand of intermediate goods by industry Z_{ci}^D , demand by household H_c^D , demand for capital creation (investment) I_c^D , demand by government G_c^D and demand for export E_c .

$$Q_c^D = \sum_i Z_{ci}^D + H_c^D + I_c^D + G_c^D + E_c$$

Demand for imported commodities:

$$Q_c^M = \sum_i Z_{ci}^M + H_c^M + I_c^M + G_c^M$$

Equation for Aggregation of Total Demand: MINIMAL

Tablo equations for the following:

$$Q_c^D = \sum_i Z_{ci}^D + H_c^D + I_c^D + G_c^D + E_c$$

$$Q_c^M = \sum_i Z_{ci}^M + H_c^M + I_c^M + G_c^M$$

is:

Equation E_x0

```
(all,c,COM)(all,s,SRC) SALES(c,s)*x0(c,s)=  
sum{u,USER,USE(c,s,u)*x(c,s,u)};
```


Equation for Aggregation of Total Demand: ORANI-G

Tablo equations for the following:

$$Q_c^D = \sum_i Z_{ci}^D + H_c^D + \sum_i I_{ci}^D + G_c^D + E_c$$

$$Q_c^M = \sum_i Z_{ci}^M + H_c^M + \sum_i I_{ci}^M + G_c^M$$

Equation

```
E_delSaleA (all,c,COM)(all,s,SRC) delSale(c,s,"Interm") =  
    0.01*sum{i,IND,V1BAS(c,s,i)*x1(c,s,i)};  
E_delSaleB (all,c,COM)(all,s,SRC) delSale(c,s,"Invest") =  
    0.01*sum{i,IND,V2BAS(c,s,i)*x2(c,s,i)};  
E_delSaleC (all,c,COM)(all,s,SRC) delSale(c,s,"HouseH")=0.01*V3BAS(c,s)*x3(c,s);  
E_delSaleD (all,c,COM) delSale(c,"dom","Export")=0.01*V4BAS(c)*x4(c);  
E_delSaleE (all,c,COM) delSale(c,"imp","Export")= 0;  
E_delSaleF (all,c,COM)(all,s,SRC) delSale(c,s,"GovGE") =0.01*V5BAS(c,s)*x5(c,s);  
E_delSaleG (all,c,COM)(all,s,SRC) delSale(c,s,"Stocks") = LEVP0(c,s)*delx6(c,s);
```

Domestic-Import Sourcing: MINIMAL & ORANI-G

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Entity that buys both domestic and imported goods behave by minimizing their combined (local and import) expenditure subject to CES combination of domestic and imported goods (composite of differentiated goods, non-infinity elasticity of substitution), or:

$$\text{minimize } P^D Q^D + P^M Q^M \text{ subject to } Q(Q^D, Q^M; \sigma_{DM})$$

The (linearized) solution:

$$q^D = q - \sigma_{DM} (p^D - p), \quad q^M = q - \sigma_{DM} (p^M - p) \quad \text{and} \quad p = S^D p^D + S^M p^M$$

σ_{DM} is called **Armington elasticities**.

Domestic-Import Sourcing Equations: MINIMAL

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$$q^D = q - \sigma_{DM} (p^D - p), \quad q^M = q - \sigma_{DM} (p^M - p) \quad \text{and} \quad p = S^D p^D + S^M p^M$$

Equation E_x

```
(all,c,COM)(all,s,SRC)(all,u,IMPUSER)
  x(c,s,u) = x_s(c,u) - SIGMA(c)*[p(c,s) - p_s(c,u)];
```

Equation E_p_s

```
(all,c,COM)(all,u,IMPUSER) p_s(c,u) =
  sum{s,SRC, SRCSHR(c,s,u)*p(c,s)};
```

Domestic-Import Sourcing Equations: ORANI-G

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```
Equation E_x1 # Source-specific commodity demands #
(all,c,COM)(all,s,SRC)(all,i,IND)
  x1(c,s,i)-a1(c,s,i) = x1_s(c,i) -SIGMA1(c)*[p1(c,s,i) +a1(c,s,i) -p1_s(c,i)];

Equation E_p1_s # Effective price of commodity composite #
(all,c,COM)(all,i,IND)
  p1_s(c,i) = sum{s,SRC, S1(c,s,i)*[p1(c,s,i) + a1(c,s,i)]};

Equation E_x2 # Source-specific commodity demands #
(all,c,COM)(all,s,SRC)(all,i,IND)
  x2(c,s,i)-a2(c,s,i) - x2_s(c,i) = - SIGMA2(c)*[p2(c,s,i)+a2(c,s,i) - p2_s(c,i)];

Equation E_p2_s # Effective price of commodity composite #
(all,c,COM)(all,i,IND)
  p2_s(c,i) = sum{s,SRC, S2(c,s,i)*[p2(c,s,i)+a2(c,s,i)]};

Equation E_x3 # Source-specific commodity demands #
(all,c,COM)(all,s,SRC)
  x3(c,s)-a3(c,s) = x3_s(c) - SIGMA3(c)*[ p3(c,s)+a3(c,s) - p3_s(c) ];

Equation E_p3_s # Effective price of commodity composite #
(all,c,COM) p3_s(c) = sum{s,SRC, S3(c,s)*[p3(c,s)+a3(c,s)]};
```

Demand for Composite Commodities and Exports: MINIMAL

- Demand of intermediate from industries is Leontief: $z_{ci} = x_i$
- Demand of commodities by households is Cobb-Douglas:

$$h_c = y - p_c$$

- Demand for investment and government is **exogenous** or $i_c = 0$ and $g_c = 0$
- Demand for exports:

$$e_c = f_c^E - \epsilon_c \left((p_c^D - \pi) - p_c^W \right)$$

where f_c^E is export shifter, ϵ_c is export elasticity, π is exchange rate, and p_c^W is international price.

Cobb-Douglas Demand System in 2x2 and MINIMAL

$$Q_i = \alpha_i \frac{Y}{P_i} \text{ for } i = 1, \dots, n$$

or in linearized form:

$$q_i = y - p_i$$

Properties:

- Unitary income elasticity: $\frac{\partial Q_i}{\partial Y} \frac{Y}{Q_i} = 1$
- Unitary own-price elasticity: $\frac{\partial Q_i}{\partial P_i} \frac{P_i}{Q_i} = -1$
- Zero cross-price elasticity: $\frac{\partial Q_i}{\partial P_j} \frac{P_j}{Q_i} = 0$
- Constant budget share: $\frac{P_i Q_i}{Y} = \alpha_i$

Stone-Geary utility (or LES) function: ORANI-G

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$$U = \prod_i (Q_i - \gamma_i)^{\beta_i}$$

- β_i is the marginal budget share and $\sum_i \beta_i = 1$
- γ_i is called the subsistence consumption level.
- $Q_i - \gamma_i$ is called supernumary consumption or luxurious consumption, the consumption after subsistence consumption
- Note that when $\gamma_i = 0$ the Stone-Geary utility function is collapsed to Cobb-Douglas
- See handout for derivation of LES Demand function

Some feature of the LES demand function:

$$Q_i = \gamma_i + \frac{\beta_i}{P_i} \left(Y - \sum_j P_j \gamma_j \right)$$

- Income elasticity: $\eta_i = \frac{\partial Q_i}{\partial Y} \frac{Y}{Q_i} = \frac{\beta_i}{P_i} \frac{Y}{Q_i} = \frac{\beta_i}{P_i} \frac{Y}{Q_i} = \frac{\beta_i}{w_i}$ where $w_i = \frac{P_i Q_i}{Y}$ is the budget share.
- Own price elasticity: $\varepsilon_{ii} = \frac{\gamma_i(1-\beta_i)}{x_i} - 1$
- Cross-price elasticity: $\varepsilon_{ij} = -\frac{\beta_i \gamma_j}{x_j} \frac{w_j}{w_i}$
- Frisch parameter: $\Phi = -\frac{Y}{Y - \sum_j P_j \gamma_j}$, is the elasticity of marginal utility of income.

LES Household Demand in ORANI-G Tablo File

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```
E_x3sub # Subsistence demand for composite commodities #
(all,c,COM) x3sub(c) = q + a3sub(c);
E_x3lux # Luxury demand for composite commodities #
(all,c,COM) x3lux(c) + p3_s(c) = w3lux + a3lux(c);
E_x3_s # Total household demand for composite commodities #
(all,c,COM) x3_s(c) = B3LUX(c)*x3lux(c) + [1-B3LUX(c)]*x3sub(c);
E_utility # Change in utility disregarding taste change terms #
utility + q = sum{c,COM, S3LUX(c)*x3lux(c)};
E_a3lux # Default setting for luxury taste shifter #
(all,c,COM) a3lux(c) = a3sub(c) - sum{k,COM, S3LUX(k)*a3sub(k)};
E_a3sub # Default setting for subsistence taste shifter #
(all,c,COM) a3sub(c) = a3_s(c) - sum{k,COM, S3_S(k)*a3_s(k)};
E_w3tot # Household budget constraint: determines w3lux #
w3tot = x3tot + p3tot;
```

Equations for Demand for Composite Goods: MINIMAL

In Tablo file:

Equation E_x1 # demand for commodity composites #

$(all, c, COM)(all, i, IND) \quad x_s(c, i) = x1tot(i);$

Equation E_x3

$(all, c, COM) \quad x_s(c, "Households") + p_s(c, "Households") = w3tot;$

Equation E_x4a $(all, c, COM) \quad x(c, "dom", "Exports") =$

$f4q(c) - ABS[EXP_ELAST(c)] * [p(c, "dom") - phi] - pworld(c);$

In CMF/Simulation file (Closure):

Exogenous $x_s(COM, "Investment") ;$

Exogenous $x_s(COM, "Government") ;$

Investment Demand: ORANI-G

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Total demand

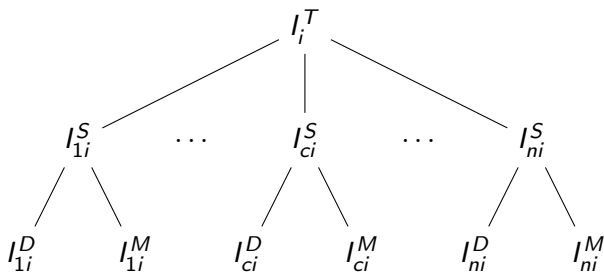
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$$I_i^T = I_i^T \left(\mathbf{I}_{ci}^S \left(\frac{I_{ci}^D}{A_{ci}^D}, \frac{I_{ci}^M}{A_{ci}^M}; \sigma^{ARM}, A_{ci}^S \right); 0, A_i^T \right)$$



Investment Demand Equation

$$I_i^T \left(\mathbf{I}_{ci}^S \left(\frac{I_{ci}^D}{A_{ci}^D}, \frac{I_{ci}^M}{A_{ci}^M}; \sigma^{ARM}, A_{ci}^S \right); 0, A_i^T \right)$$

With optimization and linearization:

- $i_{ci}^D - a_{ci}^D = i_{ci}^S - \sigma^{ARM} (p_{ci}^D + a_{ci}^D - p_{ci}^S)$
- $i_{ci}^M - a_{ci}^M = i_{ci}^S - \sigma^{ARM} (p_{ci}^M + a_{ci}^M - p_{ci}^S)$
- $p_{ci}^S = S_{ci}^D (p_{ci}^D - a_{ci}^D) + S_{ci}^M (p_{ci}^M - a_{ci}^M)$
- $i_{ci}^S - (a_{ci}^S + a_i^T) = i_i^T$
- $p_i^T = \sum_c S_{ci}^I (p_{ci}^S + a_{ci}^S + a_i^T)$

Investment Demand Equation in TABLO

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```
E_x2 # Source-specific commodity demands #  
(all,c,COM)(all,s,SRC)(all,i,IND)  
x2(c,s,i)-a2(c,s,i) - x2_s(c,i) = - SIGMA2(c)*[p2(c,s,i)+a2(c,s,i) - p2_s(c,i)];  
  
E_p2_s # Effective price of commodity composite #  
(all,c,COM)(all,i,IND)  
p2_s(c,i) = sum{s, SRC, S2(c,s,i)*[p2(c,s,i)+a2(c,s,i)]};  
  
E_x2_s (all,c,COM)(all,i,IND) x2_s(c,i) - [a2_s(c,i) + a2tot(i)] = x2tot(i);  
  
E_p2tot (all,i,IND) p2tot(i)  
= sum{c, COM, (V2PUR_S(c,i)/ID01[V2TOT(i)])*[p2_s(c,i) + a2_s(c,i) + a2tot(i)]};
```

Investment allocation across industries: I_i^T

Investment in each industry can follow one of the following rules/behaviour.

- 1 Investment in one industry follows the profitability of the industry. Profitability is measured by the gross rate of returns of the industry calculated as the ratio of the rental of one unit capital to the price of acquiring new unit of capital (price of investment per unit), or:

$$\frac{I_i}{K_i} = F \left(\frac{R_i}{P_i^I} \right)$$

- 2 Investment in one industry follows the aggregate (macro-wide) investment. In the case where aggregate investment is exogenous, the investment in that industry will also be exogenous.
- 3 Investment to capital ratio $\frac{I_i}{K_i}$ is exogenous (called long-run investment rule).

Investment allocation equation in TABLO

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Variable

```
(all,i,IND) ggro(i) # Gross growth rate of capital = Investment/capital #;  
(all,i,IND) gret(i) # Gross rate of return = Rental/[Price of new capital] #;  
(all,i,IND) finv1(i) # Shifter to enforce DPSV investment rule #;  
(all,i,IND) finv2(i) # Shifter for "exogenous" investment rule #;  
(all,i,IND) finv3(i) # Shifter for Longrun investment rule #;  
invslack # Investment slack variable for exogenizing aggregate investment #;
```

Equation

```
E_ggro (all,i,IND) ggro(i) = x2tot(i) - x1cap(i);  
E_gret (all,i,IND) gret(i) = p1cap(i) - p2tot(i);
```

Equation E_finv1 # DPSV investment rule

```
(all,i,IND) ggro(i) = finv1(i) + 0.33*[2.0*gret(i) - invslack];
```

Equation E_finv2 # Alternative rule for "exogenous" investment industries

```
(all,i,IND) x2tot(i) = x2tot_i + finv2(i);
```

Equation E_finv3 # Alternative Long-run investment rule

```
(all,i,IND) ggro(i) = finv3(i) + invslack;
```

Demand for Composite Commodities and Exports: ORANI-G

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- Demand of intermediate from industries is Leontief:

$$z_{ci} - \left(a_{ci}^Z + a_i^X \right) = x_i$$

- Demand for government is **exogenous** or follow household consumption.
- Demand for exports:

$$e_c = f_c^E - \epsilon_c \left(\left(p_c^D - \pi \right) - p_c^W \right)$$

where f_c^E is export shifter, ϵ_c is export elasticity, π is exchange rate, and p_c^W is international price.

Other composite demand in TABLO

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```
Equation E_x1_s # Demands for commodity composites #
(all,c,COM)(all,i,IND) x1_s(c,i) - [a1_s(c,i) + a1tot(i)] = x1tot(i);

E_x5 # Government demands # (all,c,COM)(all,s,SRC) x5(c,s) = f5(c,s) + f5tot;
E_f5tot # Overall government demands shift # f5tot = x3tot + f5tot2;

E_pf4 (all,c,COM) pf4(c) = p4(c) - phi;
E_x4A # Individual export demand functions #
(all,c,TRADEXP) x4(c) - f4q(c) = -ABS[EXP_ELAST(c)]*[pf4(c) - f4p(c)];

Equation E_X4B # Collective export demand functions #
(all,c,NTRADEXP) x4(c) - f4q(c) = x4_ntrad;
```

Zero Profit and Market Clearing Conditions: MINIMAL

Zero profit conditions for industries:

$$P_i^P X_i = \sum_c P_c^D Z_{ci}^D + \sum_c P_c^M Z_{ci}^M + WL_i + R_i K_i$$

Market clearing conditions for domestic commodities:

$$Q_c^D = X_c$$

Tablo equations:

Equation `E_p1tot` # *cost of production = cost of all inputs* #
(all,i,IND) `V1TOT(i)*[p1tot(i)+ x1tot(i)] =`
`sum{c,COM,sum{s,SRC, USE(c,s,i)*[p(c,s) + x(c,s,i)]}}`
`+ FACTOR("Labour",i)*[p1lab + x1lab(i)]`
`+ FACTOR("Capital",i)*[p1cap(i)+ x1cap(i)];`

Equation `E_x1tot` (all,c,COM) `x1tot(c) = x0(c,"dom");`

Commodities Market Clearing Equations: ORANI-G

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Equation

```
E_delSaleA (all,c,COM)(all,s,SRC) delSale(c,s,"Interm") =
    0.01*sum{i,IND,V1BAS(c,s,i)*x1(c,s,i)};
E_delSaleB (all,c,COM)(all,s,SRC) delSale(c,s,"Invest") =
    0.01*sum{i,IND,V2BAS(c,s,i)*x2(c,s,i)};
E_delSaleC (all,c,COM)(all,s,SRC) delSale(c,s,"HouseH")=0.01*V3BAS(c,s)*x3(c,s);
E_delSaleD (all,c,COM)          delSale(c,"dom","Export")=0.01*V4BAS(c)*x4(c);
E_delSaleE (all,c,COM)          delSale(c,"imp","Export")= 0;
E_delSaleF (all,c,COM)(all,s,SRC) delSale(c,s,"GovGE") =0.01*V5BAS(c,s)*x5(c,s);
E_delSaleG (all,c,COM)(all,s,SRC) delSale(c,s,"Stocks") = LEVP0(c,s)*delx6(c,s);
```

Equation E_p0A # Supply = Demand for domestic commodities #

```
(all,c,COM) 0.01*[TINY+DOMSALES(c)]*x0dom(c) =sum{u,LOCUSER,delSale(c,"dom",u)};
```

Variable (all,c,COM) x0imp(c) # Total supplies of imported goods #;

Equation E_x0imp # Import volumes #

```
(all,c,COM) 0.01*[TINY+V0IMP(c)]*x0imp(c) = sum{u,LOCUSER,delSale(c,"imp",u)};
```

Factor market clearing: ORANI-G

Labor market clearing:

$$L_o^S = \sum_i L_{io}$$

Equation

```
E_x1lab_i # Demand equals supply for labour of each skill #  
(all,o,OCC) V1LAB_I(o)*x1lab_i(o) = sum{i,IND, V1LAB(i,o)*x1lab(i,o)};
```

```
E_p1lab # Flexible setting of money wages #  
(all,i,IND)(all,o,OCC)  
p1lab(i,o)= p3tot + f1lab_io + f1lab_o(i) + f1lab_i(o) + f1lab(i,o);
```

```
Variable (all,o,OCC) p1lab_i(o) # Average wage of occupation #;
```

```
Equation E_p1lab_i # Average wage of occupation #  
(all,o,OCC) V1LAB_I(o)*p1lab_i(o) = sum{i,IND, V1LAB(i,o)*p1lab(i,o)};
```

Zero Profit Conditions Equation: ORANI-G

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Equation

```
E_delV1CST (all,i,IND) delV1CST(i) = delV1PRIM(i) +  
    sum{c,COM,sum{s,SRC, 0.01*V1PUR(c,s,i)*[p1(c,s,i) + x1(c,s,i)]}}  
    + 0.01*V1OCT(i) * [p1oct(i) + x1oct(i)];
```

```
E_delV1PTX (all,i,IND) delV1PTX(i) =  
    PTXRATE(i)*delV1CST(i) + V1CST(i) * delPTXRATE(i);
```

```
E_delV1TOT (all,i,IND) delV1TOT(i) = delV1CST(i) + delV1PTX(i);
```

```
E_p1tot (all,i,IND) V1TOT(i)*[p1tot(i) + x1tot(i)] = 100*delV1TOT(i);
```

Producer's Price, Purchaser's Price and Taxes

Define T_c^D and T_c^M are advalorem sales tax rate that link producer's price to purchaser's price for domestic and imported commodities respectively:

$$\text{Domestic commodities} \Rightarrow P_c^D = \left(1 + \frac{T_c^D}{100}\right) P_c^P$$

$$\text{Imported commodities} \Rightarrow P_c^M = \left(1 + \frac{T_c^M}{100}\right) P_c^W$$

Tablo equations:

Equation E_pA

(all,c,COM)

$$p(c, "dom") = p1tot(c) + 100 * [V1TOT(c) / (V1TOT(c) + V1PTX(c))] * Delptxrate(c);$$

Equation E_pB

(all,c,COM)

$$p(c, "imp") = pworld(c) + phi + 100 * [V0CIF(c) / SALES(c, "imp")] * Delmtxrate(c);$$



Closures: MINIMAL

Short-run:

```
phi           ! 1 Exchange rate $A/$US
x_s(COM,"Investment") ! COM Investment demands
x_s(COM,"Government") ! COM Government demands
x1cap         ! IND Current capital stocks
realwage      ! 1 Real wage
x3tot         ! 1 Real household consumption
a1prim        ! IND Factor-using technical change
pworld        ! COM World prices $US
f4q           ! COM Export demand shift
Delmtxrate    ! COM Tax rate on imports
Delptrrate    ! COM Tax rate on production
```

Long-run:

```
swap x1cap = gret;      ! Capital is mobile across sectors
swap realwage = employ; ! Aggregate employment is fixed
swap x3tot = delB;      ! Trade balance (per GDP) is fixed
```

Short-run and Long-run Closures

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Technical changes are exogenous: $\mathbf{a1cap}_i, \mathbf{a1lab}_{oi}, \mathbf{a1lnd}_i, \mathbf{a1prim}_i, \mathbf{a1tot}_i, \mathbf{a2tot}_i$

Land and capital are sector-specific: $\mathbf{x1lnd}_i, \mathbf{x1cap}_i \Leftrightarrow \mathbf{gret}_i$

Wage follows CPI: $\mathbf{realwage} \Leftrightarrow \mathbf{employ}_i$

Rate of returns are sector-specific: $\mathbf{capslack}$

Expenditure side of GDP are constrained:

$\mathbf{x3tot} \Leftrightarrow \mathbf{delB}, \mathbf{x2tot}_i \Leftrightarrow \mathbf{invslack}, \mathbf{x5tot} \Leftrightarrow \mathbf{f5tot2}, \mathbf{f5}_{cs}, \mathbf{delx6}_c$

Tax rates are exogenous: $\mathbf{delPTXRATE}_i, \mathbf{f0tax}_{sc}, \mathbf{f1oct}_i, \mathbf{f1tax}_{csi},$

$\mathbf{f2tax}_{csi}, \mathbf{f3tax}_{cs}, \mathbf{f4tax}_{ntrad}, \mathbf{f4tax}_{trad}, \mathbf{f5tax}_{cs}, \mathbf{t0imp}_c$

Some invest. follows profit: $\mathbf{finv1}_i \Leftrightarrow \mathbf{finv3}_i$, Some follows aggregate: $\mathbf{finv2}_i$

Import price and export demand curve are fixed:

$\mathbf{pf0cif}_c, \mathbf{f4p}_c, \mathbf{f4q}_c, \mathbf{f4p}_{ntrad}, \mathbf{f4q}_{ntrad}$

Preferences are exogenous: $\mathbf{q}, \mathbf{a3}_{sc}$ and exchange rate is numeraire: $\mathbf{phi}$

Appendix: Consumer optimization with Stone-Geary utility function (1)

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Consumer maximizes $U = \prod_i (X_i - \gamma_i)^{\beta_i}$ subject to $Y = \sum_i P_i X_i$

The Lagrange equation: $\mathcal{L} = \prod_i (X_i - \gamma_i)^{\beta_i} + \lambda (Y - \sum_i P_i X_i)$

First order conditions:

$$\beta_i (X_i - \gamma_i)^{\beta_i - 1} \prod_{j \neq i} (X_j - \gamma_j)^{\beta_j} = \lambda P_i \Rightarrow \frac{\beta_i U}{X_i - \gamma_i} = \lambda P_i \text{ and } Y = \sum_i P_i X_i$$

$$\frac{\beta_i (X_j - \gamma_j)}{\beta_j (X_i - \gamma_i)} = \frac{P_i}{P_j} \Rightarrow P_j \beta_i X_j - P_j \beta_i \gamma_j = P_i \beta_j (X_i - \gamma_i)$$

$$P_j X_j = \frac{P_i \beta_j (X_i - \gamma_i) + P_j \beta_i \gamma_j}{\beta_i}$$

Appendix: Consumer optimization with Stone-Geary utility function (2)

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$$Y = \sum_j P_j X_j = \sum_j \frac{P_i \beta_j (X_i - \gamma_i) + P_j \beta_i \gamma_j}{\beta_i} = \sum_j \frac{\beta_j P_i X_i - \beta_j P_i \gamma_i + \beta_i P_j \gamma_j}{\beta_i}$$

$$Y = \frac{\sum_j \beta_j P_i X_i - \sum_j \beta_j P_i \gamma_i + \sum_j \beta_i P_j \gamma_j}{\beta_i}$$

$$Y = \frac{P_i X_i \sum_j \beta_j - P_i \gamma_i \sum_j \beta_j + \beta_i \sum_j P_j \gamma_j}{\beta_i} = \frac{P_i X_i - P_i \gamma_i}{\beta_i} + \sum_j P_j \gamma_j$$

Appendix: Consumer optimization with Stone-Geary utility function (3)

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$$\frac{P_i X_i - P_i \gamma_i}{\beta_i} = Y - \sum_j P_j \gamma_j \Rightarrow P_i X_i - P_i \gamma_i = \beta_i \left(Y - \sum_j P_j \gamma_j \right)$$

$$P_i X_i = P_i \gamma_i + \beta_i \left(Y - \sum_j P_j \gamma_j \right) \text{ in expenditure form}$$

$$X_i = \gamma_i + \frac{\beta_i}{P_i} \left(Y - \sum_j P_j \gamma_j \right) \text{ in demand form}$$